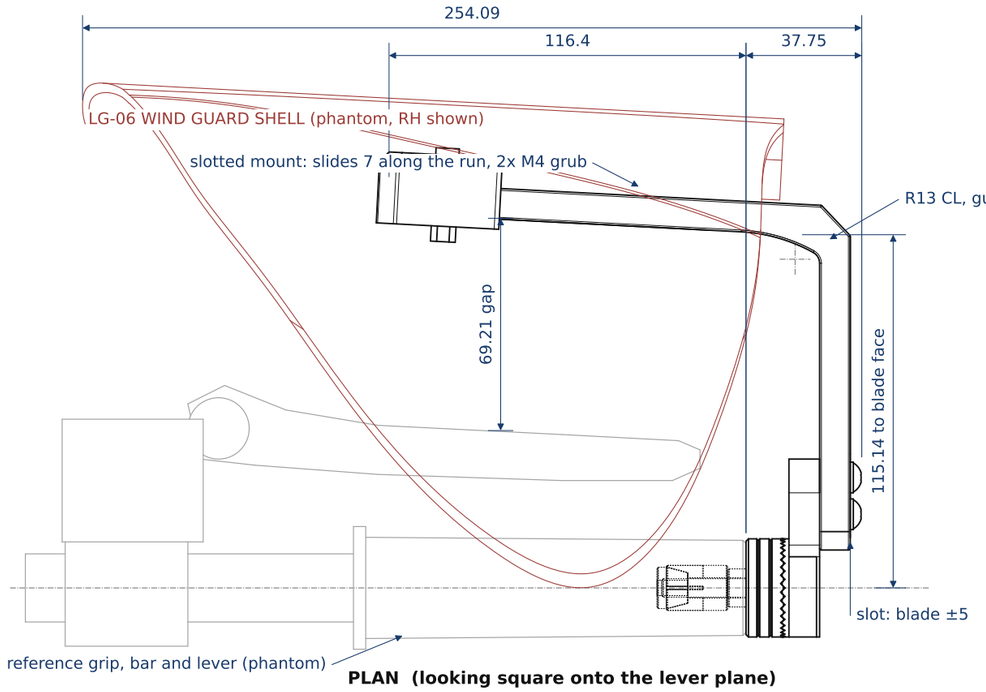
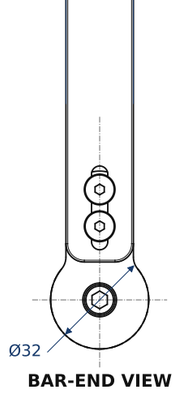
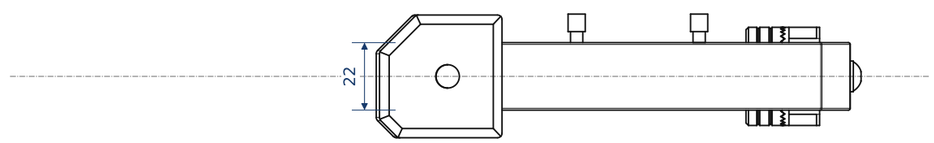


PLAN (looking square onto the lever plane)



BAR-END VIEW



FRONT VIEW (wind guard shell not shown)

NOTES

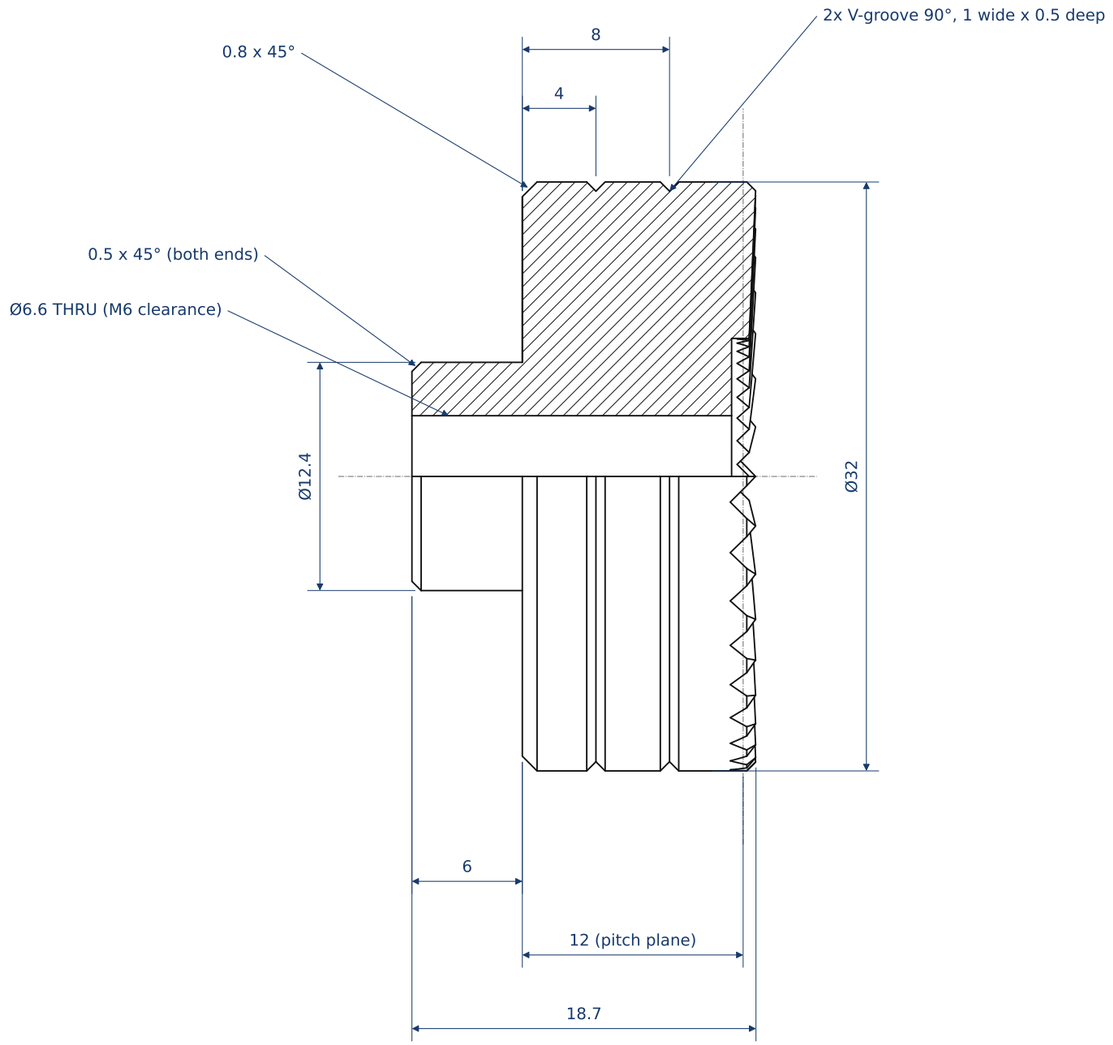
1. Drawing plane = lever plane. Turn the arm on its 36-tooth serration (10° steps) to put the blade on the lever line.
2. Clearances shown for the reference sport-bike lever; keep ≥ 10 mm all round, re-check at full lever travel.
3. Mass per side ≈ 184 g, 296 g with the wind guard.
4. Left side: same parts, arm turned over (they are symmetric about the lever plane), except the wind guard shell: LG-06 is made as a mirror-image pair.
5. Wind guard shell drawn as a phantom in the plan; its profile and fixing are on sheet 6.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

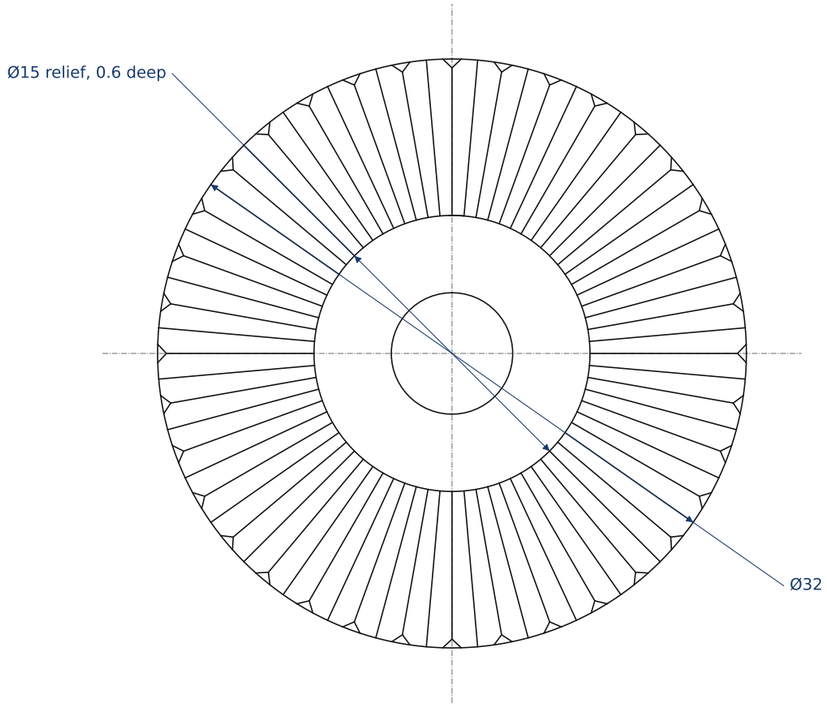
GENERAL ARRANGEMENT

LG-00

MATERIAL				FINISH	
see part sheets				see part sheets	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
1 : 2	mm	1 set	ISO 2768-mK		1 / 6
SOURCE				DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py				2026-10-23	0.2 - 0.5



HALF SECTION A-A



VIEW ON SERRATION (from outboard)

NOTES

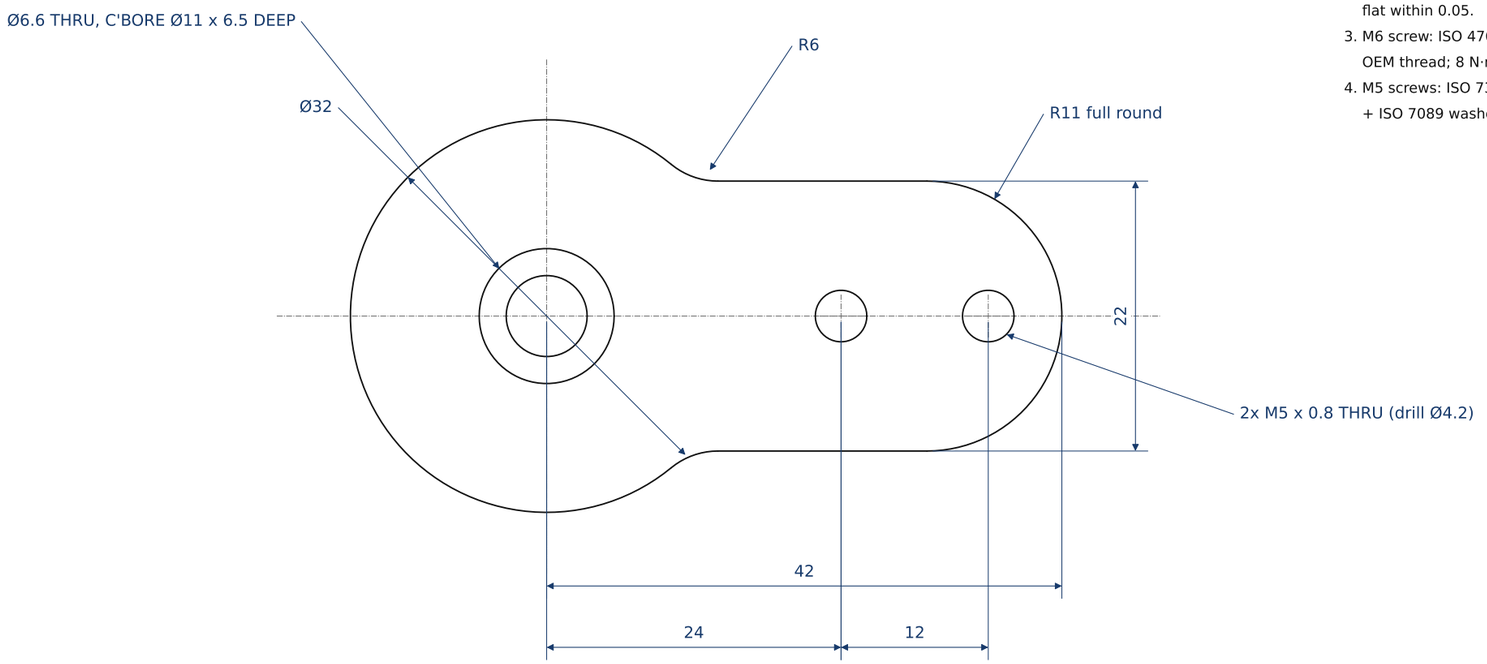
1. Hirth face serration: 36 teeth, 90° included, crests and roots radial through the axis, pitch plane at 12 from the bar-end face, toothed from Ø15 to Ø32: tooth height 1.4 at Ø32. Mates with LG-02 (offset half a tooth).
2. Variant LG-01b: same part without the spigot, for bars with a threaded OEM bar-end insert.
3. Spigot Ø12.4 centres in any bar with ID ≥ 12.5; the expander (LG-04/05) sits behind it.
4. Hard anodise after machining; mask nothing. Serration may be cut with a 90° V-cutter.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

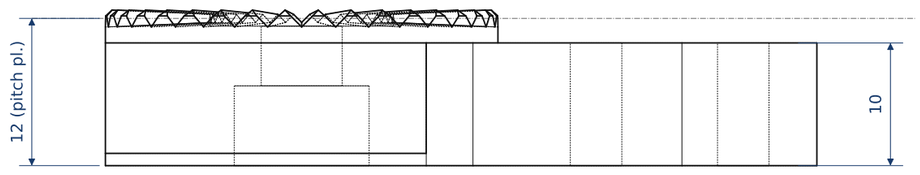
BAR-END ADAPTER

LG-01

MATERIAL			FINISH		
EN AW-6061-T6			Hard anodise, colour to order		
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
3 : 1	mm	1	ISO 2768-mK		2 / 6
SOURCE			DATE		BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py			2026-10-23		0.2 - 0.5



OUTBOARD FACE



PLAN (lever plane), serration toward the bar

NOTES

1. Hirth serration as LG-01: 36 teeth, 90°, pitch plane on the hub's inner face. Sits half a tooth out of phase with LG-01 so crests meet roots: 10° indexing.
2. Neck edges R1. Outboard face is the blade seat: flat within 0.05.
3. M6 screw: ISO 4762 M6 x 45 (expander) or to suit the OEM thread; 8 N·m, medium threadlocker.
4. M5 screws: ISO 7380 M5 x 14 (x 16 with LG-03b) + ISO 7089 washers, 4 N·m.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

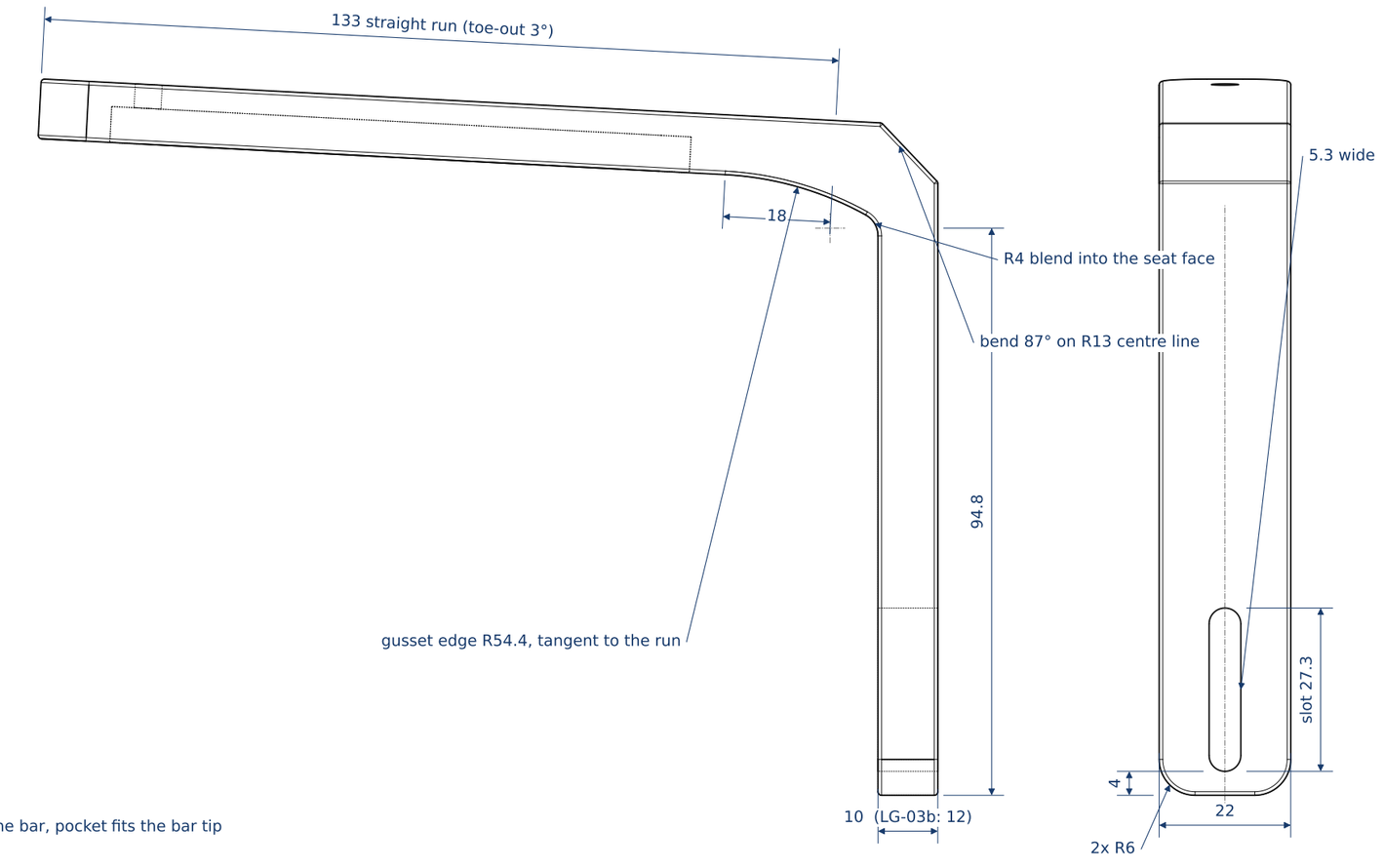
GUARD ARM

LG-02

MATERIAL				FINISH	
EN AW-6061-T6				Hard anodise, colour to order	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
2 : 1	mm	1	ISO 2768-mK		3 / 6
SOURCE				DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py				2026-10-23	0.2 - 0.5

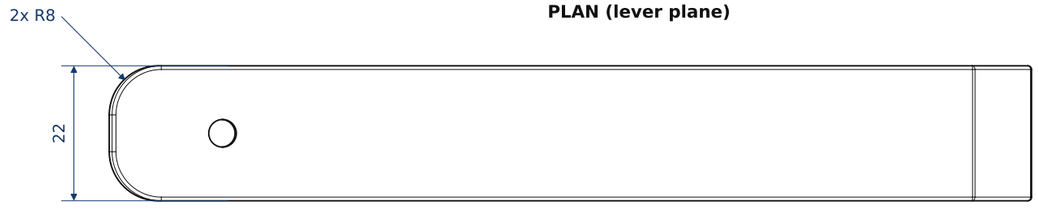
NOTES

1. Centre-line length 247.5 mm: 94.8 straight + 87° bend on R13 + 133 straight.
2. Gusset inside the bend: section 10 -> 13 mm across the bend, about 1.7x the bending strength where a fall loads the blade.
   Its edge is tangent all along (blend into the seat face, arc, run-out on the run): no corner to start a crack.
3. Shape after the reference bar (photos): long leg ~150 mm, short leg ~120 mm to the outer corner,
   faceted outer corner, channel 6 deep in the run's lever side (3 mm flanges).
4. LG-03 10 thick: injection-moulded PA6-GF30 or PP-GF, or printed PA12-CF / PETG-CF lying flat
   (lever plane on the bed, channel up), 6 walls, 50 % gyroid.
5. LG-03b heavy duty: same outline, 12 thick, gusset run-out 24 mm, printed as LG-03.
6. Slot lets the blade slide ±5 mm toward / away from the lever.

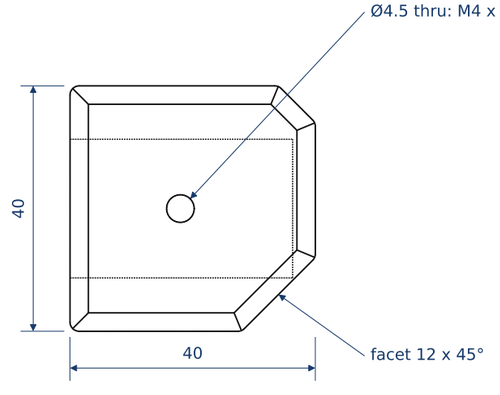


PLAN (lever plane)

REAR PLATE (outboard)



FRONT VIEW (outer face)

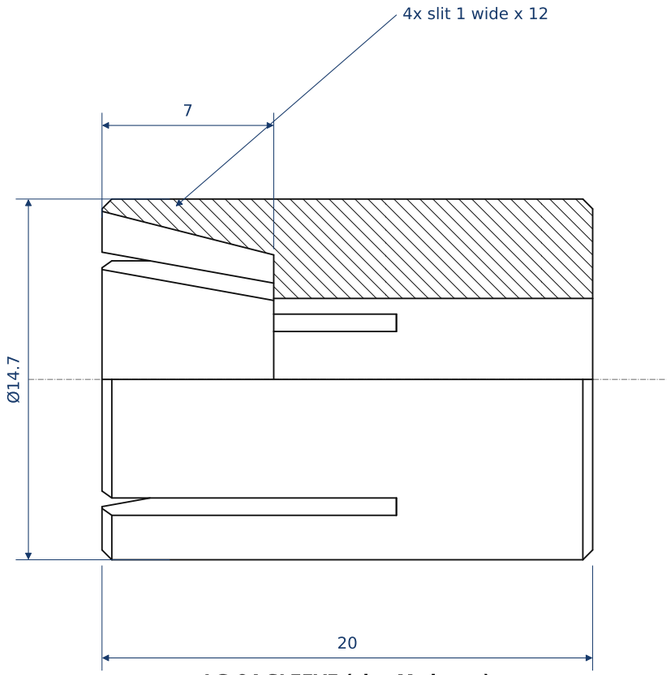


LG-08 BAR TIP GUARD, outer face (1:1)

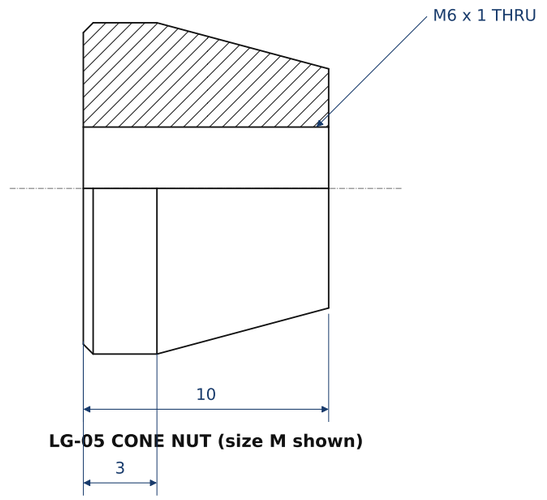
RACING LEVER GUARD / UNIVERSAL BAR-END FIT

GUARD BAR (L) WITH BEND GUSSET — LG-03

MATERIAL				FINISH	
PA6-GF30 / PP-GF moulded; PA12-CF / PETG-CF printed				Black, edges R0.6	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
1 : 1	mm	1	ISO 2768-mK		4 / 6
SOURCE				DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py				2026-10-23	0.2 - 0.5



LG-04 SLEEVE (size M shown)



LG-05 CONE NUT (size M shown)

SIZES (all lengths as drawn: sleeve 20, recess 7, cone land 3)

Size	Bar ID	Sleeve OD	Recess Ø face / bottom	Cone Ø big / small	Taper (half)
S	13 - 15	12.7	11.7 / 8.4	11.5 / 8	14°
M	15 - 17	14.7	13.7 / 10.15	13.5 / 9.75	15°
L	17 - 19	16.7	15.7 / 12.15	15.5 / 11.75	15°

NOTES

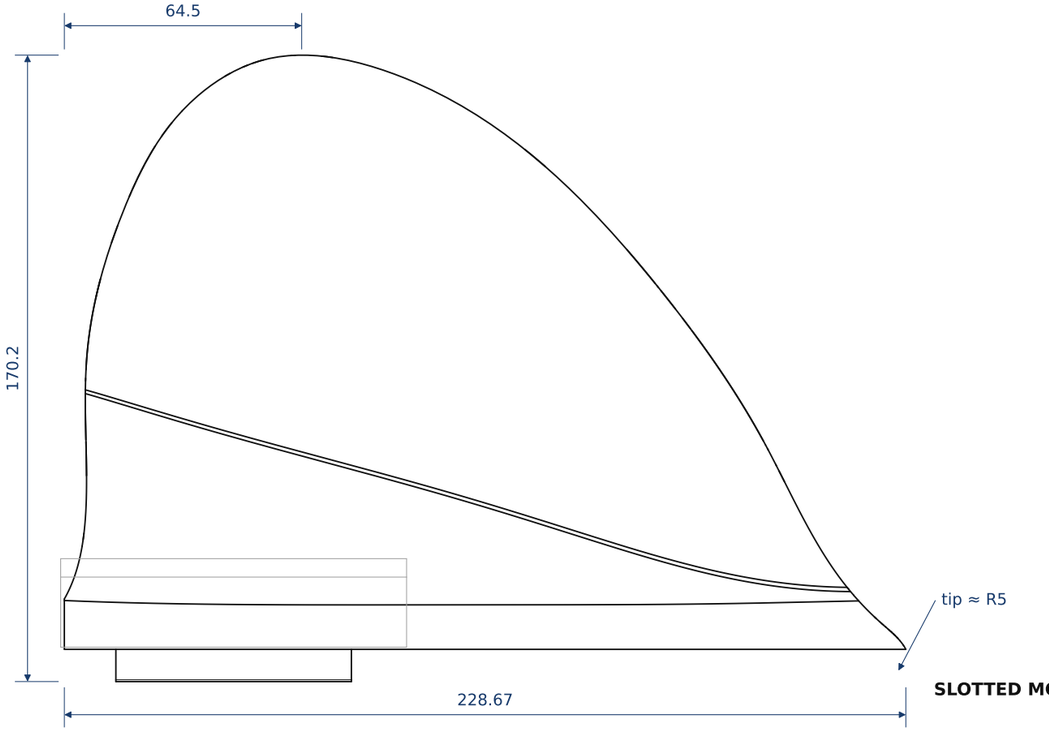
1. Bolt chain: M6 x 45 (ISO 4762) through LG-02 and LG-01 into the cone nut: 8.5 mm thread engagement at fit-up.
2. Tightening pulls the cone into the slit sleeve, which grips the bar bore. 8 N·m. Hold the arm square on its serration while tightening.
3. Pick the size from the bar's measured inside diameter.
4. Sleeve can be printed in TPU 95A (100 % infill) or cut from NBR tube; cone may be steel for higher load.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

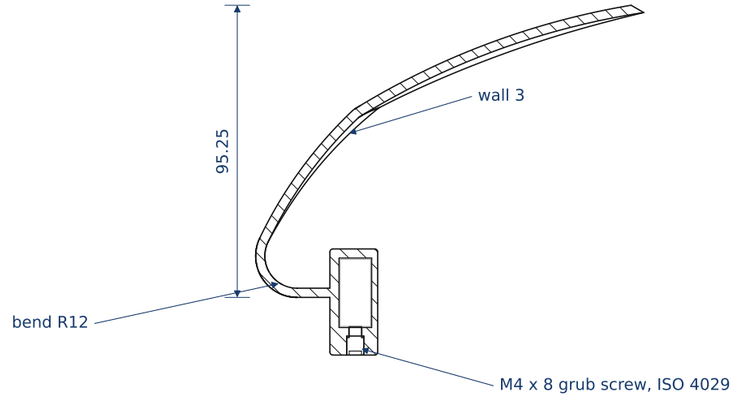
EXPANDER SET

LG-04 / LG-05

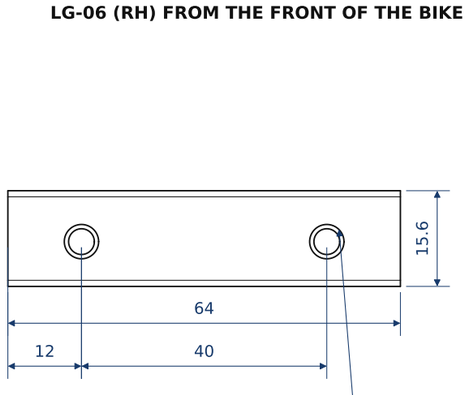
MATERIAL: LG-04 TPU 95A or NBR 70A; LG-05 6061-T6				FINISH: LG-05 clear anodise or zinc	
SCALE: 4 : 1	UNITS: mm	QTY / SIDE: 1 sleeve + 1 cone	GENERAL TOLERANCES: ISO 2768-mK	PROJECTION	SHEET: 5 / 6
SOURCE: cad/lever_guard.py (CadQuery), drawings.py				DATE: 2026-10-23	BREAK SHARP EDGES: 0.2 - 0.5



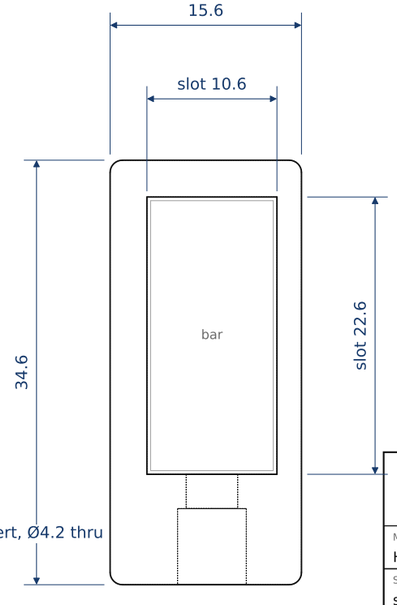
LG-06 (RH) FROM THE FRONT OF THE BIKE (1:1.67) - bar phantom



SECTION THROUGH A GRUB SCREW (1:2)



SLOTTED MOUNT FROM BELOW



SLOTTED MOUNT, END VIEW (2:1)

NOTES

1. LG-06 surface as the STEP files 06 (RH) / 06b (LH, mirror image). One piece: shield, flange and 64 mm slotted mount on a full-length pad. Moulded HDPE / PP, or printed ASA / PA12 standing on the pad (shield up, tree supports under the bend and the top edge). R1 edges.
2. Press two M4 x 5.7 brass heat-set inserts into the pad (iron at about 230 °C), flush with the pad face.
3. Slide the mount onto the bar's long leg from its tip (before the tip guard LG-08), shield in front of the lever.
4. Lock with 2x M4 x 8 grub screws (ISO 4029, cup point) from below onto the bar's lower edge, 1 N·m. Slides 19-26 mm from the bend.
5. Check steering lock, mirrors and reservoir before riding.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

WIND GUARD WITH SLOTTED MOUNT

LG-06

MATERIAL					FINISH	
HDPE or PP, moulded; ASA / PA12 printed					colour to order, edges R1	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET	
see views	mm	1 pair (RH + LH)	ISO 2768-mK		6 / 6	
SOURCE					DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py					2026-10-23	0.2 - 0.5